Overview of the OpenQSE Compiler Effort

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OpenQSE Day, July 24 2026

Welcome!

Welcome to the compiler working group!

Goals for today:

Begin a Survey of Quantum Compiler Technologies

Develop Shared Terminology

Identify Key Gaps

Agree upon Working Group Goals, Milestones, and Deliverables

Gather initial Compiler Working Group participants for ongoing work

Have something to share?
Sign up for a lightning talk!
Lightning talks will be early in the PM session!

Scope & Focus: What this workstream is responsible for

The compilation layers in the openQSE reference architecture including definition of OpenQSE-preferred circuit formats and providing for translation among circuit formats in order to connect SDKs to QPUs in a portable manner.

In-scope:

Specify an OpenQSE middleware compiler component, including interfaces to software from other workstreams:

- Quantum Resource Interface (QRI) interactions with the compiler (device queries, interaction with device compiler, circuit interchange format)
- Workflow/Application (System Architecture and Workflows WG) interactions with compiler (SDK adapters)
- Runtime layer (SW Arch WG) interactions with the compiler, if any

Develop reference implementations of such a compiler

Adapters that connect quantum SDKs to the OpenQSE compiler

Circuit compilation and translation that supports the above

Making choices as to preferred compiler IR and circuit interchange formats

Key Problems

How to provide a consistent level of functionality for hybrid quantum-classical circuits across different vendors systems and quantum SDKs

How to use compilation and translation methods to handle widely varying circuit input formats from different vendor systems

How to integrate with existing compilation steps in SDKs and in vendor stacks?

How to support FTQC compilation?

What compilation steps should SDKs provide? What compilation steps should QPU devices provide?

Where in the architecture should compilation steps run?

Alignment with OpenQSE Objective

Interoperability. Support the necessary circuit translation for portability across SDKs and QPU providers.

Vendor-neutral interfaces. Specify what the OpenQSE compiler component does to allow for multiple implementations. Provide necessary functionality to connect circuits written in SDKs to circuits run by QPUs.

HPC–QC integration. Enable compilation steps on nearby HPC resources. Enable compilation of circuits with mid-circuit classical compute on HPC. Enable HPC-oriented quantum SDKs.

Expected Outputs (End of Day)

Milestones (12 months)

Milestone 1 (6 months):

Draft survey paper

Milestone 2 (12 months):

Draft specification of interfaces

Prototype for reference implementation

Deliverables (12 months)

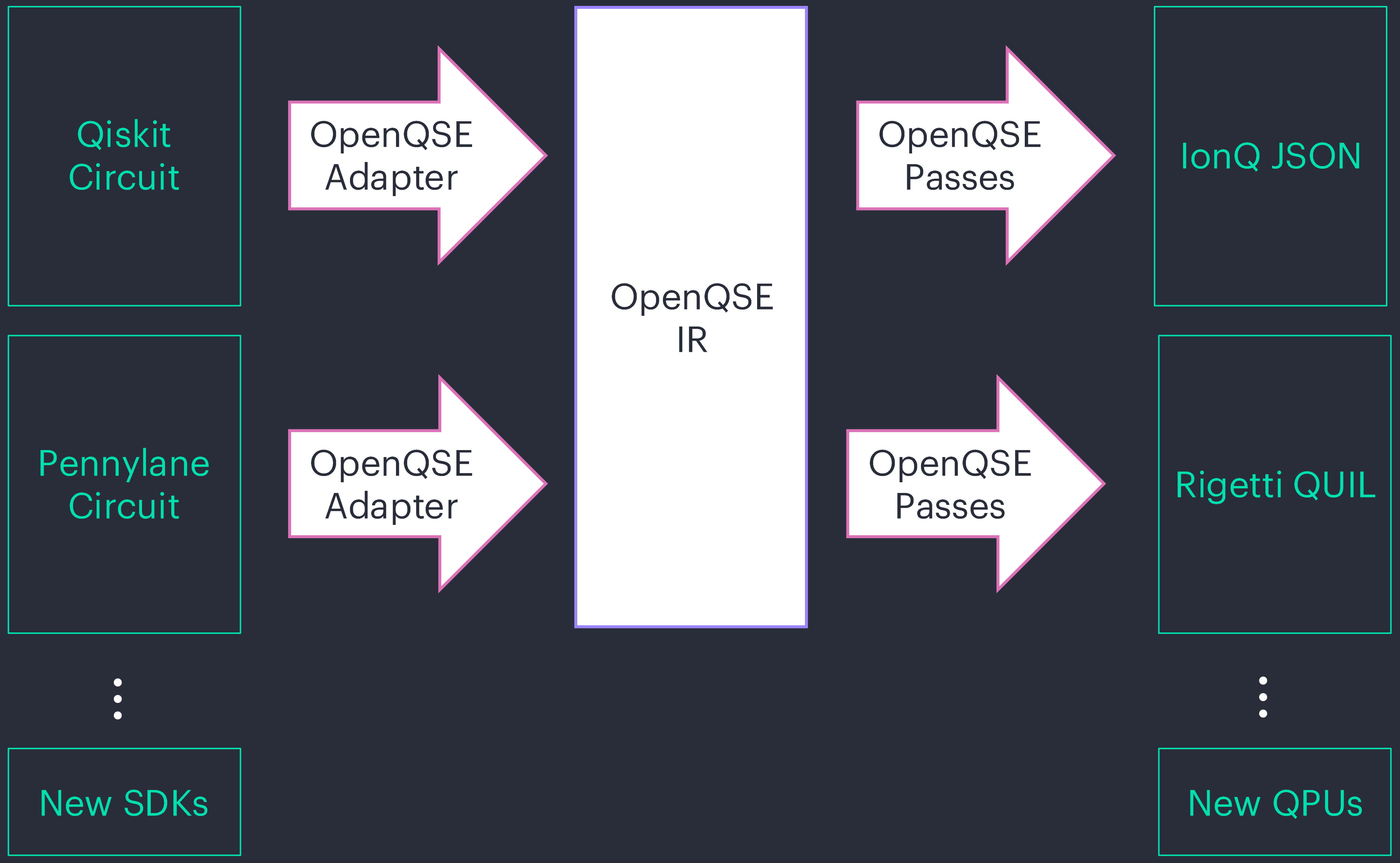
Survey paper

- Understand existing quantum compilers; what IRs they use; what passes they perform
- Define criteria to select an IR
- Include a chart about quantum-classical capabilities
- Is there convergence on IRs? Testing?
- Performance and capabilities in terms of circuit size; how do tools scale with qubit and gate count?

Compiler-related requirements for architecture document

Specification of OpenQSE compiler component

Reference implementation


Qiskit Circuit
OpenQSE Adapter
Pennylane Circuit
OpenQSE Adapter
New SDKs
OpenQSE IR
OpenQSE Passes
IonQ JSON
OpenQSE Passes
Rigetti QUIL
New QPUs

Probably handles:
- Specifying circuits

May handle:
- Compiling to native gates
- User-controllable transformation passes

Probably handles:
- Translation between circuit interchange formats

May handle:
- Compiling to native gates
- Mapping
- QEC compilation
- User-controllable transformations

Probably handles:
- Compiling to native gates
- Mapping
    - (with calibration data)
- QPU-specific optimization

May handle:
- QEC compilation
    - (optimized for QPU)

The Role of the OpenQSE Compiler

I am proposing two roles for the OpenQSE compiler effort:

1. Provide circuit translation needed for portability & specify interfaces
   - (e.g., translating from OpenQASM 3 to QUIL, ...)
   - Connect circuit compilers in SDKs to QPU-specific circuit compilers

2. Create a community around compiler passes that can be reused by different vendors

Schedule

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Session 1 (AM) Schedule ~~9:45~~ 10:00-12:00

9:45-10:00	**Overview of the OpenQSE Compiler Effort**
10:00-10:15	**Introductions**
10:15-10:45	**Background: Classical Compilers, Quantum Circuit Compilers, FTQC Compilers**
10:45-11:00	Break (chat with your neighbors!)
11:00-12:00	**Discussion: Capabilities of Existing Quantum Compilers & Terminology**

Session 2 (PM) Schedule 2:15-4:15

2:15-2:25	**Introductions**
2:30-2:50	**Lightning Talks**
2:50-3:20	**Discussion: Key Gaps & Goals: What does an OpenQSE compiler need to do?**
3:20-3:25	Break (chat with your neighbors!)
3:25-4:00	**Open Discussion & Process for Next Steps**
4:00-4:15	**Prepare Report-Out**